General Science (Tech) — Class 9 Solved Exercises — PCTB

Class 9 — Solved Exercises

Punjab Curriculum and Textbook Board (PCTB)

6 chapters — 132 exercises

Chapter 1: Introduction and Role of Science

Exercise 1.1 — Fill in the blanks

Jabir Bin Hayyan is the founder of _____.

chemistry

Exercise 1.1 — Fill in the blanks

The study of animals is called _____.

zoology

Exercise 1.1 — Fill in the blanks

Bu Ali Sina is called _____ of the Muslim world.

Aristotle

Exercise 1.1 — Fill in the blanks

_____ divided chemical compounds into four categories.

Al-Razi (Muhammad Bin Zikrya Al-Razi)

Exercise 1.1 — Fill in the blanks

Kitab-ul-Manazir is the first book on _____.

light (optics)

Exercise 1.2 — True / False

Bu Ali Sina is considered to be one of the founders of medicine.

True

Exercise 1.2 — True / False

Jabir was first to discuss the causes, symptoms, cure and prevention of smallpox and measles.

False

Exercise 1.2 — True / False

Jabir was expert in physics.

False

Exercise 1.2 — True / False

Kitab-ul-Manazir is a publication of Al-Bairuni.

False

Exercise 1.2 — True / False

The knowledge about animals is called botany.

False

Exercise 1.3 — Multiple Choice Questions

Ibn-ul-Haitham was related to which branch of science?

(a) Chemistry

(b) Biology

(c) Light ✓

(d) Geology

Correct answer: Light

Exercise 1.3 — Multiple Choice Questions

Which is the famous book by Al-Bairuni?

(a) Al-Qanoon-fil-Tib

(b) Kitab-ul-Manazir

(c) Kitab-ul-Hawi

(d) Tahreer-ul-Amakin ✓

Correct answer: Tahreer-ul-Amakin

Exercise 1.3 — Multiple Choice Questions

Which branch of science is related to mechanics, heat, light and sound?

(a) Chemistry

(b) Biology

(c) Astronomy

(d) Physics ✓

Correct answer: Physics

Exercise 1.3 — Multiple Choice Questions

Kahuta is renamed as a tribute to which scientist?

(a) Dr. Abdus Salam

(b) Dr. Abdul Qadeer Khan ✓

(c) Dr. Atta-ur-Rehman

(d) Dr. Samar Mubarak Mand

Correct answer: Dr. Abdul Qadeer Khan

Exercise 1.3 — Multiple Choice Questions

Which Pakistani scientist won the Nobel Prize?

(a) Dr. Abdul Qadeer Khan

(b) Dr. Munir Ahmad Khan

(c) Dr. Abdus Salam ✓

(d) Dr. Atta-ur-Rehman

Correct answer: Dr. Abdus Salam

Exercise 1.4

Q4. What is meant by science?

Science comes from the Latin word Scientia, which means to study and analyse facts in their original form. Its basic principle is observation and hypothesis, and a scientific law is established in the light of experiments. In short, science is the systematic study of facts through observation, hypothesis and experiment.

Exercise 1.5

Q5. What are the five branches of science?

The five main branches given in the chapter are: (1) Physics — the study of matter, energy and motion, including mechanics, heat, light and sound. (2) Chemistry — the study of the composition and properties of matter. (3) Biology — the study of living things, plants and animals. (4) Astronomy — the study of the universe, stars and planets. (5) Mathematics — the study of numbers, quantities and shapes, on which the other sciences depend. The chapter also lists Agriculture, Medicine and Geography as further branches.

Exercise 1.6

Q6. Write down the names of two Muslim scientists and their contributions.

1) Jabir Bin Hayyan — the founder of chemistry. He extracted metals, prepared nitric, sulphuric and hydrochloric acids, and discovered the properties of sulphur, mercury and sodium carbonate. 2) Muhammad Bin Zikrya Al-Razi — a chemist and physician. He was the first to explain the causes, symptoms, treatment and prevention of chicken pox and smallpox, and he classified chemical substances into four groups: plants, animals, minerals and man-made substances.

Exercise 1.7

Q7. Write down the names of two Pakistani scientists and their contributions.

1) Dr. Abdus Salam — the only Pakistani Nobel laureate (1979). He gave the theory of the unification of the weak nuclear force and the electromagnetic

force, and founded the International Centre for Theoretical Physics at Trieste. 2) Dr. Abdul Qadeer Khan — the founder of Pakistan's nuclear programme. The Kahuta research laboratories were renamed Dr. A. Q. Khan Research Laboratories in his honour, and Pakistan conducted its nuclear tests at Chaghi on 28 May 1998 under his work.

Exercise 1.8

Q8. Explain the importance of science in the light of the Holy Quran.

The Holy Quran repeatedly invites man to observe and think: Allah says that if the ocean were ink for the words of the Lord, the ocean would be exhausted before His words (Surah Al-Kahf 18:109), and that man has been given only a little knowledge (Surah Bani Israel 17:85). The first revelation commands, 'Read! In the name of your Lord' (Surah Al-Alaq 96:1-5), and Allah says He created everything in pairs (Surah Al-Zariyat 51:49). The Holy Prophet (PBUH) said seeking knowledge is an obligation on every man and woman, and to seek knowledge from the cradle to the grave. Science is therefore the study of Allah's creation through observation and reason.

Chapter 2: Our Life and Chemistry

Exercise 2.1 — Fill in the blanks

The process by which green plants manufacture carbohydrates (glucose) from carbon dioxide and water is called _____.

photosynthesis

Exercise 2.1 — Fill in the blanks

Natural gas contains _____ percent methane.

NOT STATED IN CHAPTER 2 — the chapter only says natural gas is methane (p. 13); no percentage is given, so none is guessed here.

Exercise 2.1 — Fill in the blanks

_____ is the only chemical compound found naturally as a liquid, as a solid (ice) and as a gas (water vapours).

Water

Exercise 2.1 — Fill in the blanks

Nitrogen occurs in plants and animals in the form of _____.

proteins

Exercise 2.1 — Fill in the blanks

A dilute solution of iodine in ethanol is known as _____.

tincture of iodine

Exercise 2.1 — Fill in the blanks

Phosphorus is an important constituent of _____.

ATP and ADP (the energy-storage phosphate compounds)

Exercise 2.2 — True / False

Hydrogen is the most abundant element in living things.

False

Exercise 2.2 — True / False

The photosynthesis process is called a catabolic process.

False

Exercise 2.2 — True / False

Most of our food is obtained from carbohydrates.

True

Exercise 2.2 — True / False

Two thirds of the human body is made up of water.

True

Exercise 2.2 — True / False

Ice is lighter than water.

True

Exercise 2.3 — Multiple Choice Questions

What is the percentage of nitrogen in the air?

(a) 78% ✓

(b) 21%

(c) 0.9%

(d) 0.03%

Correct answer: 78%

Exercise 2.3 — Multiple Choice Questions

Natural gas contains what percentage of methane? (Note: no percentage is stated in Chapter 2; the chapter only identifies natural gas as methane.)

No percentage is stated in Chapter 2. The chapter only identifies natural gas as methane (p. 13: 'natural gas (methane)'). The four options given in the book are 70%, 80%, 85% and 95%, but none of them is supported by the chapter text, so no option is marked correct here rather than guessing one.

Exercise 2.3 — Multiple Choice Questions

Which is the most abundant element in the human body?

(a) Oxygen ✓

(b) Hydrogen

(c) Carbon

(d) Nitrogen

Correct answer: Oxygen

Exercise 2.3 — Multiple Choice Questions

Which of the following is an essential element? (Note: the chapter also lists chlorine as essential, section 2.5 viii.)

(a) Chlorine

(b) Copper

(c) Sodium ✓

(d) Bromine

Correct answer: Sodium

Exercise 2.3 — Multiple Choice Questions

What is the percentage of sodium in leaf tissue?

(a) 0.01 to 10% ✓

(b) 1 to 4%

(c) 0.2 to 1%

(d) 0.1 to 0.4%

Correct answer: 0.01 to 10%

Exercise 2.3 — Multiple Choice Questions

What is the percentage of oxygen in the air?

(a) 78%

(b) 21% ✓

(c) 0.9%

(d) 0.03%

Correct answer: 21%

Exercise 2.3 — Multiple Choice Questions

What is the percentage of water in tomatoes?

(a) 87%

(b) 86%

(c) 84%

(d) 95% ✓

Correct answer: 95%

Exercise 2.3 — Multiple Choice Questions

Which is not a function of water?

(a) Universal solvent

(b) For cooking

(c) Photosynthesis

(d) Soluble ✓

Correct answer: Soluble

Exercise 2.4

Q4. (i) What do you know about carbon compounds as a source of energy?

Carbon compounds are the main source of energy for living things. Carbohydrates are the simplest food compounds, and glucose is their simplest form. In respiration, glucose is oxidised with oxygen to give carbon dioxide, water and energy: Glucose + Oxygen -> Carbon dioxide + Water + Energy. This energy runs all the activities of the body.

Exercise 2.4

Q4. (ii) What is the anomalous expansion of water?

Water does not behave like other liquids on cooling. Its density is maximum at 4 degrees Celsius. When water below 4 degrees is cooled further, it expands instead of contracting, so that at 0 degrees the density of ice is only 0.918 g/cm3 while water is about 1.00 g/cm3. Ice is therefore lighter than water and floats, and the water below the ice stays at 4 degrees, which protects fish and other aquatic animals in winter.

Exercise 2.4

Q4. (iii) What is photosynthesis?

Photosynthesis is the process by which green plants manufacture carbohydrates (glucose) from atmospheric carbon dioxide and water from the soil, in the presence of sunlight energy and chlorophyll: Water + Carbon dioxide -> Glucose + Oxygen. It takes place only in the chlorophyll-containing cells of leaves and stems, oxygen is released as a by-product, and it is an anabolic (building-up) process, the reverse of respiration.

Exercise 2.4

Q4. (iv) What do you know about chlorine?

Chlorine is a very poisonous gas, but it has many beneficial uses in daily life. It is used to kill germs and bacteria, in domestic bleaches, and in water purification and swimming pools. Polyvinyl chloride (PVC), a common plastic compound of chlorine, is used as an insulator and waterproofing material. Chlorine is also essential for higher plants and mammals: sodium chloride acts as an electrolyte and hydrochloric acid acts as a digestive juice in the body.

Exercise 2.4

Q4. (v) What are the properties of oxygen?

Oxygen is a colourless, odourless gas which is slightly soluble in water, so fish and aquatic life depend on it. It is the second most abundant gas in air after nitrogen (21% by volume) and is essential for life. It supports respiration, in which food (mainly glucose) is oxidised to release energy, and combustion, for which fuel, heat and oxygen are all required. Ozone, produced from oxygen in the atmosphere, protects living things by absorbing the sun's ultraviolet light.

Exercise 2.4

Q4. (vi) What do you know about nitrogen?

Nitrogen occurs in the atmosphere as diatomic molecules and is its major constituent (78% by volume). It is relatively inert and acts as an important diluent of air, slowing down combustion and corrosion. Nitrogen also occurs in plants and animals in the form of proteins: plants make protein from nitrates in the soil, and animals obtain protein by eating plants. Through the nitrogen cycle, nitrogen moves regularly between living organisms, soil and air, keeping its amount in air constant.

Chapter 3: Biochemistry and Biotechnology

Exercise 3.1 — Fill in the blanks

Vaccination stimulates the _____ system of the body.

immune

Exercise 3.1 — Fill in the blanks

Human blood can be classified into A, B, AB and O groups on the basis of antigen and _____.

antibody

Exercise 3.1 — Fill in the blanks

Enzymes are very _____ in their function.

specific

Exercise 3.2 — True / False

Metabolism is a combination of anabolic and catabolic reaction.

True

Exercise 3.2 — True / False

The term 'Biotechnology' was introduced in 1980.

False

Exercise 3.3 — Multiple Choice Questions

The function of platelets is

(a) Blood Clotting 

(b) Engulfing the bacteria

(c) To produce antibiotics

(d) Transfer of oxygen

Correct answer: Blood Clotting

Exercise 3.3 — Multiple Choice Questions

Enzyme Papain is obtained from (note: the book prints two options labelled (b); the correct one is Papaya plant)

(a) Rose plant

(b) Carrot

(c) Papaya plant ✓

(d) Wheat

Correct answer: Papaya plant

Exercise 3.3 — Multiple Choice Questions

Which blood group is called universal recipient?

(a) A

(b) B

(c) O

(d) AB ✓

Correct answer: AB

Exercise 3.4

Q4. (i) Write the names of three main types of cells in blood.

The three main types of cells in blood are: (1) Red blood cells (erythrocytes), which carry oxygen. (2) White blood cells (leucocytes), which defend the body against germs. (3) Platelets (thrombocytes), which help the blood to clot.

Exercise 3.4

Q4. (ii) What is transgenic organism?

A transgenic organism is an organism whose genetic material has been changed

by genetic engineering, by inserting a gene from another organism into it. The new gene gives the organism a desired character, for example Bt cotton, which carries a bacterial gene that protects it from insect pests.

Exercise 3.4

Q4. (iii) What is a catalyst?

A catalyst is a substance that speeds up a chemical reaction without itself being used up or changed at the end of the reaction. Enzymes are biological catalysts: they speed up the chemical reactions inside living bodies but remain unchanged themselves.

Exercise 3.5

Q5. What is metabolism?

Metabolism is the sum total of all the chemical reactions taking place in a living body. It has two parts: anabolic reactions, which build up substances (like photosynthesis), and catabolic reactions, which break down substances to release energy (like respiration).

Exercise 3.6

Q6. What is an enzyme? What role is played by enzymes in our daily life?

An enzyme is a biological catalyst, usually a protein, made by living cells, which speeds up a biochemical reaction without being used up itself, and each enzyme is very specific in its function. In daily life enzymes digest our food (for example amylase digests starch and pepsin digests proteins), papain from papaya tenderises meat, and enzymes are used in making bread and cheese and in washing powders to remove stains.

Exercise 3.7

Q7. What are the components of blood?

Blood has two main parts. The liquid part is plasma, which carries dissolved substances around the body. Floating in the plasma are the blood cells: red blood cells for carrying oxygen, white blood cells for fighting germs, and platelets for clotting the blood.

Exercise 3.8

Q8. What is genetic engineering? How can it help in improving agriculture and livestock?

Genetic engineering is the technique of changing an organism's genes by inserting, removing or modifying a gene to give it a desired character. In agriculture it produces high-yield, disease-resistant and pest-resistant crops, such as Bt cotton. In livestock it can improve milk and meat production and breed animals resistant to disease, raising both quantity and quality.

Exercise 3.9

Q9. What is an antibiotic?

An antibiotic is a medicine, originally produced by a microorganism, which kills bacteria or stops their growth. It is used to treat bacterial infections, for example penicillin, the first antibiotic, is used against many bacterial diseases.

Exercise 3.10

Q10. What is recycling? Give a detailed account how the waste and scarce materials can be made useable.

Recycling means collecting waste materials and processing them to make new, useful products instead of throwing them away. Its benefits, given in the chapter, are: (1) less rubbish, so less environmental pollution; (2) conservation of raw materials and natural resources; (3) treatment of sewage saves water, which is

scarce in many parts of the world; (4) saving of energy and money. Paper, cardboard, plastic, rubber, metal and glass discarded as rubbish are retrieved and recycled through their industries: reusing newspapers saves trees and prevents deforestation, broken glass is crushed to make new containers, and aluminium cans and bottle caps are recycled to save materials, energy and money. Sewage is treated in sewage tanks and the purified water is discharged into rivers and reservoirs for reuse. Household paper waste can be burnt to provide heat and hot water, and organic waste gives manure and heat energy. The three methods of waste disposal in developed countries are compost (natural fertilizer), burning in incinerators, and burying underground by hygienic methods.

Chapter 4: Human Health

Exercise 4.1 — Fill in the blanks

The most abundant naturally occurring carbohydrate of the world is _____.

cellulose

Exercise 4.1 — Fill in the blanks

Fats and oils are formed by chemical combination of fatty acids with _____.

glycerol

Exercise 4.1 — Fill in the blanks

Deficiency of vitamin _____ causes night blindness.

A

Exercise 4.1 — Fill in the blanks

Goiter is caused by the deficiency of _____.

iodine

Exercise 4.1 — Fill in the blanks

Insulin and _____ are formed in pancreas.

glucagon

Exercise 4.1 — Fill in the blanks

Rabies is caused by _____ bite.

dog

Exercise 4.2 — True / False

Glucose is the building block of proteins.

False

Exercise 4.2 — True / False

Vitamin A is a fat soluble vitamin.

True

Exercise 4.2 — True / False

Rickets is caused by vitamin C deficiency.

False

Exercise 4.2 — True / False

Thyroxin hormone is secreted by parathyroid gland.

False

Exercise 4.3 — Multiple Choice Questions

The compound which is required in very small amount is

(a) Carbohydrate

(b) Proteins

(c) Fats

(d) Vitamins ✓

Correct answer: Vitamins

Exercise 4.3 — Multiple Choice Questions

The amount of energy gained from one gram of fats is

(a) 9.3 Kcal ✓

(b) 18 Kcal

(c) 27 Kcal

(d) 36 Kcal

Correct answer: 9.3 Kcal

Exercise 4.3 — Multiple Choice Questions

The disease which is caused in children due to vitamin D deficiency is

(a) Scurvy

(b) T.B.

(c) Rickets ✓

(d) Anaemia

Correct answer: Rickets

Exercise 4.3 — Multiple Choice Questions

The hormone which controls involuntary action of body is

(a) Thyroxin

(b) Epinephrine ✓

(c) Adrenal

(d) Insulin

Correct answer: Epinephrine

Exercise 4.3 — Multiple Choice Questions

The disease caused by deficiency of iodine is

(a) Goiter ✓

(b) Night blindness

(c) Malaria

(d) Cough

Correct answer: Goiter

Exercise 4.4

Q4. (i) What are the basic components of food?

The basic components of food are carbohydrates, proteins, fats, vitamins, minerals and water. Roughage (fibre) from plant food is also an important part of

the diet, because it helps digestion and prevents constipation.

Exercise 4.4

Q4. (ii) What is the role of vitamin B in body?

The B-group vitamins help the body to release energy from food, keep the nerves and skin healthy, and help in the formation of red blood cells. Deficiency of vitamin B causes beri-beri and weakness of nerves, so B vitamins are essential for energy metabolism and a healthy nervous system.

Exercise 4.4

Q4. (iii) What is the role of iron in the body?

Iron is essential to all living organisms. It is present at the active centre of haemoglobin and myoglobin, the molecules that carry oxygen in the blood and muscles, and of cytochromes and ferredoxins for electron transport. Without enough iron the blood cannot carry oxygen properly and anaemia results.

Exercise 4.4

Q4. (iv) Which disease is caused by dog bite?

Rabies is caused by the bite of a rabid dog. The virus attacks the nervous system, so a dog-bite patient must be taken to hospital at once for anti-rabies treatment and vaccination.

Exercise 4.4

Q4. (v) What is the role of insulin?

Insulin is a hormone formed in the pancreas. It lowers the level of glucose in the blood by helping the body cells to take in glucose and use or store it. When insulin is deficient, blood glucose rises and the disease diabetes results.

Exercise 4.5

Q5. Write a note on important components of food.

(1) Carbohydrates are the main source of energy; glucose is their simplest form and starch and cellulose are their common forms. (2) Proteins, made of amino acids, build and repair the body. (3) Fats and oils, formed from fatty acids and glycerol, store energy and give 9.3 Kcal per gram. (4) Vitamins are needed in very small amounts: A, D, E and K are fat soluble, while B and C are water soluble. (5) Minerals such as iron, iodine, calcium, sodium and potassium are needed for blood, bones, nerves and glands. (6) Water, about two thirds of the body, acts as a solvent and carrier and regulates body temperature.

Exercise 4.6

Q6. What do you know about protein?

Proteins are the building materials of the body. They are made of small units called amino acids, and foods containing carbon, hydrogen, oxygen and nitrogen are proteins, for example meat, fish, pulses, milk and eggs. Proteins build new cells, repair worn-out tissues, and form enzymes, hormones and antibodies. The flesh of human beings, animals, birds and fishes is made of proteins.

Exercise 4.7

Q7. What are vitamins? Describe them in detail.

Vitamins are organic compounds which the body needs in very small amounts but cannot do without; they do not give energy themselves, but they regulate the body's working. They are of two kinds: fat-soluble vitamins A, D, E and K, which are stored in the body, and water-soluble vitamins B and C, which must be taken regularly in food. Vitamin A keeps the eyes healthy and its deficiency causes night blindness. Vitamin D helps bones absorb calcium and its deficiency in children causes rickets. Vitamin C prevents scurvy and keeps gums and skin

healthy. The B-group vitamins release energy from food and keep nerves healthy.

Exercise 4.8

Q8. What is balanced diet? Describe the balanced diet for various people of our society.

A balanced diet is a diet which contains all the components of food — carbohydrates, proteins, fats, vitamins, minerals, water and roughage — in the right proportions for a person's age, work and health. Growing children need more protein and calcium for growth of bones and muscles. Hard-working labourers need more carbohydrates and fats for energy. Pregnant and nursing mothers need extra protein, iron and calcium. Old people need less fat and energy but enough calcium, vitamins and fibre. Students and office workers need a moderate, balanced diet with plenty of vegetables, milk and fruit and less fried food.

Exercise 4.9

Q9. Why exercise is important for health?

Exercise keeps the body fit and healthy in many ways. It strengthens the heart and muscles, improves blood circulation and breathing, and helps digestion. It controls body weight and prevents obesity, diabetes and heart disease. It also refreshes the mind, reduces stress and improves sleep, so regular exercise is as important for health as a balanced diet.

Exercise 4.10

Q10. Discuss the role of various endocrine glands in our body.

Endocrine glands pour their hormones directly into the blood. The pituitary gland is the master gland that controls other glands. The thyroid gland secretes

thyroxin, which controls the rate of metabolism in the body. The parathyroid glands control calcium in blood and bones. The pancreas (islets of Langerhans) secretes insulin and glucagon, which control blood glucose. The adrenal glands secrete epinephrine (adrenaline), the emergency hormone that prepares the body for sudden action by quickening the heartbeat and breathing. The gonads secrete sex hormones that control growth and reproduction.

Chapter 5: Diseases, Cause and Prevention

Exercise 5.1 — Fill in the blanks

Bacteria can be seen by a _____.

microscope

Exercise 5.1 — Fill in the blanks

E.P.I. stands for _____.

Expanded Programme on Immunization

Exercise 5.1 — Fill in the blanks

The AIDS virus is called _____.

HIV (Human Immunodeficiency Virus)

Exercise 5.1 — Fill in the blanks

Hepatitis virus spreads by faeco _____ route through food and water.

oral

Exercise 5.1 — Fill in the blanks

B.C.G injection protects from _____.

tuberculosis (T.B.)

Exercise 5.2 — True / False

Polio virus affects the nervous system.

True

Exercise 5.2 — True / False

Antibiotic drugs are used against the viral diseases.

False

Exercise 5.2 — True / False

Tuberculosis is a non treatable disease.

False

Exercise 5.2 — True / False

AIDS is not an infectious disease.

False

Exercise 5.2 — True / False

Cigarette smoker remains protected from lung and heart diseases.

False

Exercise 5.3 — Multiple Choice Questions

The drinks which can be used more in hepatitis are

(a) Water

(b) Sugarcane juice

(c) Juices

(d) All ✓

Correct answer: All

Exercise 5.3 — Multiple Choice Questions

The age of the child in which he is given first injection of BCG is

(a) First month

(b) At Birth ✓

(c) Third month

(d) Ninth month

Correct answer: At Birth

Exercise 5.3 — Multiple Choice Questions

The chemical present in smoke of cigarette which makes the person addict is

(a) Tar

(b) Nicotine

(c) Carbon monoxide

(d) Nitrogen dioxide

Correct answer: Nicotine

Exercise 5.4

Q4. (i) What is the name of AIDS virus?

The AIDS virus is called HIV, the Human Immunodeficiency Virus. It attacks the body's immune system and destroys its power to fight infections, so the patient becomes prey to many diseases.

Exercise 5.4

Q4. (ii) In which diseases does DPT injection give immunity to the body?

The DPT injection gives immunity against three diseases: Diphtheria, Pertussis (whooping cough) and Tetanus. It is part of the Expanded Programme on Immunization (E.P.I.) for children.

Exercise 5.4

Q4. (iii) How does the malaria spread?

Malaria spreads through the bite of the female Anopheles mosquito. When this mosquito bites a malaria patient and then a healthy person, it injects the malaria parasite (Plasmodium) into the healthy person's blood, and the disease begins.

Exercise 5.4

Q4. (iv) Write the names of different agents causing diseases.

The different agents that cause diseases are: (1) Bacteria, for example tuberculosis and typhoid. (2) Viruses, for example polio, AIDS and hepatitis. (3) Fungi, which cause skin diseases. (4) Protozoa, for example the Plasmodium that causes malaria. (5) Worms and other parasites that live in the body.

Exercise 5.5

Q5. How does the AIDS spread? Write its preventive measures.

AIDS spreads in four main ways: transfusion of infected blood, use of unsterilised syringes, needles or razors, sexual contact with an infected person, and from an infected mother to her child during birth or breast-feeding. It does not spread by casual contact, sharing food or shaking hands. Prevention: blood must be tested before transfusion, only disposable syringes should be used, razors and needles must not be shared, and people must be educated about safe behaviour and early testing.

Exercise 5.6

Q6. Write different methods to prevent from malaria.

Malaria can be prevented by controlling the mosquito and avoiding its bite: (1) Do not let water stand anywhere, because mosquitoes breed in stagnant water. (2) Sleep under mosquito nets. (3) Spray insecticides in houses and surroundings. (4) Cover the body with clothes, especially in the evening. (5) Take anti-malarial medicine on a doctor's advice when travelling to, or living in, a malaria area, and get fever tested at once.

Exercise 5.7

Q7. What are the effects of smoke and smoking?

Cigarette smoke contains nicotine, which makes the smoker addicted, tar, which damages the lungs and causes lung cancer and bronchitis, and carbon

monoxide, which reduces the blood's power to carry oxygen. Smoking therefore causes lung and heart diseases, weakens stamina, stains teeth and fingers, and harms non-smokers too through passive smoking. Quitting smoking greatly reduces all these risks.

Exercise 5.8

Q8. Briefly describe some of the mental diseases.

Mental diseases disturb a person's thinking, mood and behaviour. Depression is a long-lasting state of sadness and loss of interest in life. Anxiety is excessive worry and fear that disturbs daily work. Schizophrenia is a serious disease in which the patient loses contact with reality and may hear voices or hold false beliefs. Drug abuse also causes mental disease: hallucinogens, for example cannabis, disrupt the brain's function to such an extent that the person is unable to recognise time, place, sound, colour and vision. Such patients need medical treatment, care and sympathy, not ridicule.

Chapter 6: Environment and Natural Resources

Exercise 6.1 — Fill in the blanks

Ozone prevents _____ from reaching the earth.

ultraviolet rays

Exercise 6.1 — Fill in the blanks

Radiation of _____ wavelength can not go out of greenhouse.

infrared

Exercise 6.1 — Fill in the blanks

Substances that cause pollution of the environment are called _____.

pollutants

Exercise 6.1 — Fill in the blanks

Fossil fuels and minerals are _____ resources.

non-renewable

Exercise 6.1 — Fill in the blanks

One reason for the extinction of many species is _____ destruction.

habitat

Exercise 6.1 — Fill in the blanks

Areas reserved for wildlife are called _____.

national parks

Exercise 6.1 — Fill in the blanks

Moving out from an area and settling in another area is called _____.

migration

Exercise 6.2 — Multiple Choice Questions

The proportion of carbon dioxide in air is (note: Chapter 2, Table 2.2 gives 0.03%; 0.04% is the nearest option in this chapter's list)

(a) 40 percent

(b) 0.4 percents

(c) 0.04 percent 

(d) 0.004 percent

Correct answer: 0.04 percent

Exercise 6.2 — Multiple Choice Questions

Major cause of depletion of ozone layer is

(a) Oxygen

(b) Hydrogen

(c) Chlorofluorocarbons 

(d) Hydrocarbon

Correct answer: Chlorofluorocarbons

Exercise 6.2 — Multiple Choice Questions

About _____ percent population of Pakistan is dependent upon agriculture

(a) 90%

(b) 80%

(c) 60% 

(d) 50%

Correct answer: 60%

Exercise 6.2 — Multiple Choice Questions

The total number of persons living in an area is called

(a) Species

(b) Population

(c) Community

(d) Habitat

Correct answer: Population

Exercise 6.3

Q3. (i) Define: (a) Pollution (b) Pollutants (c) Recycling (d) Endangered species.

(a) Pollution is the addition of harmful substances to the environment in such amounts that air, water or land becomes unfit for living things. (b) Pollutants are the substances that cause pollution, for example smoke, sewage, industrial waste and plastic. (c) Recycling means collecting waste materials and processing them to make new, useful products instead of throwing them away. (d) An endangered species is a species of plant or animal whose numbers have become so small, mainly through habitat destruction and hunting, that it is in danger of extinction.

Exercise 6.3

Q3. (ii) Write any two effects of greenhouse effect on environment.

(1) The temperature of the earth rises (global warming), because carbon dioxide and other gases trap infrared radiation. (2) The higher temperature causes the ice on the mountains to melt, which leads to flooding, and the weather pattern of the planet is also affected, with rising sea levels threatening coastal areas.

Exercise 6.3

Q3. (iii) Write two methods of conservation of natural resources.

(1) Reuse and recycle materials such as paper, glass, metal and plastic, so that fewer raw materials are used and less rubbish is produced. (2) Use resources wisely and plant trees: save water and energy, protect forests, and use

renewable sources of energy such as sunlight, wind and water instead of exhausting fossil fuels.

Exercise 6.3

Q3. (iv) Write two benefits of wildlife.

(1) Wildlife maintains the balance of nature: every animal and plant has a place in food chains, and removing one species disturbs the whole ecosystem. (2) Wildlife is useful to man: it provides food, medicines and raw materials, and its beauty and variety attract tourists and add to the wealth and pleasure of a country.

Exercise 6.4

Q4. Write a note on depletion of ozone layer.

The ozone layer in the upper atmosphere absorbs the sun's harmful ultraviolet rays and protects living things. It is being depleted mainly by chlorofluorocarbons (CFCs), chemicals used in refrigerators, air conditioners and spray cans. When CFCs reach the ozone layer they break down ozone, so more ultraviolet radiation reaches the earth. This increases skin cancer and eye damage in man and harms crops and animals. Depletion can be controlled by banning CFCs and using safe alternatives.

Exercise 6.5

Q5. What is meant by greenhouse effect? Describe its causes and effects on environment.

The greenhouse effect is the warming of the earth because gases such as carbon dioxide absorb infrared rays and do not let them escape into space, just as the glass of a greenhouse traps heat inside. Its causes are the burning of too much carbon-containing fuel in factories, vehicles and homes, which liberates too

much carbon dioxide, and the cutting of forests, which removes the trees that absorb carbon dioxide. Its effects are a rise in the earth's temperature (global warming), melting of mountain ice, flooding, changes in weather patterns and a rise in sea level.

Exercise 6.6

Q6. How human activities affect environment?

Human activities affect the environment in many harmful ways. Burning fossil fuels in factories and vehicles pollutes the air with smoke and carbon dioxide. Clearing forests to meet the needs of a growing population causes unfavourable climatic changes, soil erosion and waste of fertile agricultural land. Industrial and domestic waste pollutes rivers and land. The growing population itself creates shortages of food, water, housing and jobs, forces many people to live in slums, and increases the number of illiterate children, lowering the quality of life of the nation.

Exercise 6.7

Q7. Write down causes, effects and measures of controlling aquatic pollution.

Causes: untreated sewage, industrial waste, oil spills, and fertilisers and pesticides washed from fields into rivers and lakes. Effects: fish and other aquatic animals die, water becomes unfit for human consumption, and water-borne diseases spread. Control: sewage and industrial waste must be treated before discharge, dumping of waste into water must be banned by law, oil spills must be cleaned at once, use of harmful chemicals near water must be reduced, and people must be taught to keep water clean.

Exercise 6.8

Q8. Explain use and environmental effects of fossil fuels.

Fossil fuels — coal, petroleum and natural gas — are used to produce electricity, run factories and vehicles, cook food and heat homes; they are the main source of energy in the world today. Their environmental effects are serious: burning them liberates carbon dioxide, which causes the greenhouse effect and global warming, and smoke, which pollutes the air and causes breathing diseases. Oil spills pollute seas and coasts. Moreover, fossil fuels are non-renewable resources, so wasteful use exhausts them for future generations.

Exercise 6.9

Q9. Write a note on conservation of resources.

Conservation of resources means using natural resources wisely so that they last for future generations. It includes recycling and reusing waste materials, which saves raw materials, energy and money; saving water and electricity and avoiding waste; planting trees and protecting forests; using renewable sources of energy such as sunlight, wind and water; and protecting wildlife and soil. By conserving resources we control pollution, keep the environment healthy and secure the needs of the growing population.

Exercise 6.10

Q10. Write brief notes on: (a) Mechanized farming and modern trends in agriculture. (b) Dairy, poultry and fish farming. (c) Wildlife conservation and National parks. (d) Importance of wildlife.

(a) Mechanized farming uses tractors, harvesters, tube wells, high-yield seeds, chemical fertilisers and pesticides, and modern methods of irrigation and storage. It greatly increases production and reduces labour, but careless use of chemicals and water can harm soil and health. (b) Dairy farming is rearing cattle for milk, poultry farming is rearing hens for eggs and meat, and fish farming is rearing fish in ponds. All three provide protein-rich food, employment and

income to farmers. (c) Wildlife conservation means protecting wild animals and plants and their habitats. National parks are large areas reserved for wildlife where hunting and cutting of trees are banned, so animals can live and breed safely. (d) Wildlife is important because it maintains the balance of nature and food chains, provides food, medicines and raw materials, and adds beauty and tourist value to a country.

Exercise 6.11

Q11. Explain environmental problems arising due to increase in population.

Increase in population creates many environmental problems. More people need more food, water, housing, schools and jobs, but resources are limited, so shortages arise and many people are forced to live in slums. Forests are cleared to fulfil the needs of the growing population, which leads to unfavourable climatic changes, soil erosion and waste of fertile agricultural land. More waste and smoke increase pollution of air, water and land. The quality of life of a nation, judged by education, health, nutrition, housing, clean water and electricity, falls, and the number of illiterate children increases day by day.